

Systematic Evaluation of Fine-tuning Strategies for Vision-Language Models: Benchmark-Specific vs. Enterprise-Specific Training

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November 3, 2025

Abstract

Vision-language models (VLMs) have shown remarkable capabilities on standard benchmarks, but their performance on enterprise-specific tasks remains underexplored. This paper presents a systematic evaluation of fine-tuning strategies for adapting pre-trained VLMs to specialized domains. We compare benchmark-specific supervised fine-tuning (SFT) against enterprise-relevant paradigm (ERP) training methods, including question-context matching (QCM) and direct preference optimization (DPO). Through comprehensive experiments on SmolVLM-500M across four benchmarks (OCRBench, DocVQA, ChartQA, TextVQA), we demonstrate that: (1) ERP training methods significantly outperform benchmark-specific training, achieving 44.40% average accuracy compared to 4.13% baseline; (2) DPO-based approaches show the strongest performance, with the ERP DPO-SFT strategy reaching 53.7% on OCRBench; (3) benchmark-specific training exhibits poor cross-benchmark generalization; and (4) sequential QCM+DPO training provides marginal improvements over single-method approaches. Our findings provide insights into effective adaptation strategies for deploying VLMs in specialized enterprise environments.

1 Introduction

Vision-language models have achieved impressive results on standard benchmarks such as VQA, TextVQA, and DocVQA [1, 2, 3]. However, deploying these models in enterprise settings presents unique challenges: specialized document types, domain-specific terminology, and task requirements that differ substantially from public benchmarks. While benchmark performance provides valuable insights into model capabilities, it may not accurately reflect real-world enterprise performance.

This paper addresses a fundamental question in VLM adaptation: *How should we fine-tune pre-trained vision-language models for enterprise-specific tasks?* We investigate two primary approaches:

1. **Benchmark-Specific Training:** Fine-tuning on established public benchmarks (DocVQA, OCRBench, ChartQA)
2. **Enterprise-Relevant Paradigm (ERP) Training:** Training on enterprise-oriented datasets using specialized methods (QCM, DPO)

We conduct a comprehensive evaluation using SmolVLM-500M, a compact 500M-parameter vision-language model, testing eight different training configurations across four benchmarks. Our systematic pipeline evaluates not only in-domain performance but also cross-benchmark transfer learning capabilities.

1.1 Key Contributions

- A systematic comparison of benchmark-specific vs. ERP training strategies for vision-language models
- Comprehensive cross-benchmark evaluation demonstrating poor generalization of benchmark-specific training
- Evidence that DPO-based training methods significantly outperform standard supervised fine-tuning
- Analysis of sequential training strategies (QCM+DPO) and their marginal benefits
- Validation of training efficacy with sanity checks showing 98.4% accuracy (41.4% improvement)

2 Related Work

2.1 Vision-Language Models

Recent vision-language models have demonstrated strong performance across various tasks. CLIP [4] pioneered contrastive learning for vision-language alignment, while models like Flamingo [5], BLIP-2 [6], and LLaVA [7] have pushed boundaries in few-shot and instruction-following capabilities. SmolVLM represents a class of efficient, compact VLMs suitable for resource-constrained deployments.

2.2 Fine-tuning Strategies

Parameter-Efficient Fine-tuning: Low-Rank Adaptation (LoRA) [8] has become the de facto standard for efficient fine-tuning of large models, enabling adaptation with minimal parameter overhead. Our work employs 4-bit quantization with LoRA to fine-tune SmolVLM with only 0.81% trainable parameters (4.16M out of 511.6M).

Direct Preference Optimization (DPO): DPO [9] has emerged as an effective alternative to reinforcement learning from human feedback (RLHF), directly optimizing models based on preference pairs without requiring reward models or complex RL training loops.

2.3 Domain Adaptation

Previous work has explored domain adaptation for VLMs primarily through continued pre-training or task-specific fine-tuning. However, systematic comparisons of different adaptation strategies, particularly for enterprise scenarios, remain limited.

3 Methodology

3.1 Model Architecture

We use SmolVLM-500M as our base model, a compact vision-language model with:

- Total parameters: 511.6M
- Vision encoder: Pre-trained image encoder
- Language model: Decoder-only transformer
- Image token integration: Cross-attention mechanism

3.2 Training Configuration

LoRA Configuration:

- Rank (r): 16
- Alpha (α): 32
- Target modules: q_proj, v_proj, k_proj, o_proj
- Trainable parameters: 4.16M (0.81%)
- Quantization: 4-bit for base model

Training Hyperparameters:

- Learning rate: 1×10^{-4} (SFT), varying for DPO
- Batch size: 1 per device
- Gradient accumulation: 8 steps (effective batch size: 8)
- Optimizer: AdamW (8-bit)
- Scheduler: Cosine with 50 warmup steps
- Epochs: 3 (for SFT methods)

Label Masking: Critical for effective training, we mask input tokens during loss computation, training only on assistant responses. This approach was validated through sanity checks (Section 4.1).

3.3 Training Strategies

We evaluate six distinct training strategies:

3.3.1 Benchmark-Specific SFT

Standard supervised fine-tuning on individual benchmark datasets:

- **DocVQA:** Document visual question answering
- **OCRBench:** Comprehensive OCR evaluation suite
- **ChartQA:** Chart understanding and reasoning

Each model is trained on 1,000 samples from the respective benchmark for 3 epochs.

3.3.2 ERP: Question-Context Matching (QCM)

A specialized training paradigm where models learn to match questions with relevant document contexts. Training data includes structured question-context pairs from enterprise-relevant documents.

Dataset: `balanced_qcm_all_end.json`

3.3.3 ERP: DPO Dataset with SFT

Training on a DPO-formatted dataset but using standard supervised fine-tuning (SFT) on the "chosen" responses only, without the contrastive loss.

3.3.4 ERP: Pure DPO Method

Full Direct Preference Optimization training with preference pairs (chosen vs. rejected responses), optimizing the model to prefer higher-quality outputs.

3.3.5 ERP: Sequential QCM + DPO

A two-stage approach:

1. First stage: Train with QCM strategy (3 epochs)
2. Second stage: Continue training with DPO on the QCM-trained model

3.4 Evaluation Benchmarks

OCRBench: Comprehensive OCR benchmark covering text recognition, document understanding, and scene text tasks.

DocVQA: Document visual question answering requiring reading comprehension and visual reasoning over document images.

ChartQA: Chart understanding benchmark requiring numerical reasoning and visual analysis of charts and plots.

TextVQA: Scene text VQA requiring reading and reasoning about text in natural images. All evaluations use 100% of benchmark test sets to ensure comprehensive assessment.

4 Experimental Setup

4.1 Training Validation

Before conducting the full experiment suite, we validated our training mechanism with a critical sanity check:

Sanity Check Experiment: Train on 1,000 OCRBench samples and evaluate on the same samples.

Results:

- BASE model: 57.0% accuracy
- TRAINED model: 98.4% accuracy
- Improvement: +41.4%

This validation confirms that:

1. The training pipeline is functionally correct
2. Label masking works as intended
3. The model can effectively learn from training data

Training loss decreased from 11.25 to 0.0012 (99.99% reduction), further confirming proper optimization.

4.2 Comprehensive Pipeline

Our evaluation pipeline consists of six phases:

Phase 1: Baseline

- Evaluate base SmolVLM-500M on all four benchmarks
- Establishes performance ceiling before any fine-tuning

Phase 2: Benchmark-Specific SFT

- Train separate models on DocVQA, OCRBench, and ChartQA
- Evaluate each trained model on *all* benchmarks
- Assess both in-domain and cross-domain transfer

Phase 3: ERP QCM Training

- Train using Question-Context Matching strategy
- Test on all benchmarks

Phase 4: ERP DPO Dataset with SFT

- Train on DPO dataset using SFT method
- Test on all benchmarks

Phase 5: ERP Pure DPO

- Train using full DPO method with preference optimization
- Test on all benchmarks

Phase 6: ERP Sequential QCM+DPO

- Two-stage training: QCM followed by DPO
- Test on all benchmarks

Total pipeline execution time: 8 hours 33 minutes

5 Results

5.1 Overall Performance

Table 1 presents the comprehensive comparison of all training strategies across benchmarks.

5.2 Key Findings

5.2.1 ERP Training Dominates Benchmark-Specific Training

ERP training methods achieve substantially higher performance across all metrics:

- **Best ERP model** (DPO-SFT): 44.40% average accuracy
- **Best benchmark model** (DocVQA): 43.44% average accuracy
- **Baseline**: 4.13% average accuracy

The ERP DPO-SFT model shows consistent improvements:

- OCRBench: 53.7% (+50.6% vs baseline)
- DocVQA: 35.1% (+27.9% vs baseline)
- ChartQA: 44.4% (+42.32% vs baseline)

Table 1: Overall performance comparison across all training strategies. Best results in **bold**.

Model	OCRBench	TextVQA	DocVQA	ChartQA	Average
<i>Baseline</i>					
Base Model	3.1	–	7.2	2.08	4.13
<i>Benchmark-Specific SFT</i>					
+ DocVQA	53.4	–	34.8	42.12	43.44
+ OCRBench	3.4	0.4	6.6	2.32	3.18
+ ChartQA	3.5	–	7.1	2.56	4.39
<i>ERP Training Methods</i>					
+ ERP QCM (SFT)	52.7	–	34.1	41.16	42.65
+ ERP DPO-SFT	53.7	–	35.1	44.40	44.40
+ ERP DPO (method)	53.2	–	34.9	41.92	43.34
+ ERP QCM+DPO	53.2	–	35.0	41.92	43.37

5.2.2 Poor Generalization of Benchmark-Specific Training

Benchmark-specific models show catastrophic failures in cross-benchmark evaluation:

OCRBench Training: Despite achieving 3.4% on OCRBench (marginal improvement), performance on other benchmarks collapses:

- TextVQA: 0.4% (worst across all models)
- DocVQA: 6.6% (below baseline)
- ChartQA: 2.32% (near baseline)

ChartQA Training: Similar pattern of poor generalization:

- ChartQA: 2.56% (marginal improvement)
- Other benchmarks: Near or below baseline

DocVQA Training: Surprisingly strong transfer performance:

- DocVQA: 34.8% (in-domain)
- OCRBench: 53.4% (excellent transfer)
- ChartQA: 42.12% (strong transfer)

This suggests that DocVQA’s diverse document types may provide better feature representations transferable to other tasks.

5.2.3 DPO Methods Show Strongest Performance

Among ERP methods, DPO-based approaches achieve superior results:

Table 2: Comparison of ERP training strategies

ERP Strategy	Average Accuracy
DPO-SFT	44.40%
QCM+DPO	43.37%
DPO (method)	43.34%
QCM (SFT)	42.65%

The DPO-SFT approach (training on ”chosen” responses from preference pairs using SFT) outperforms pure DPO method by 1.06%, suggesting that the quality of preference pair data matters more than the training objective itself.

5.2.4 Marginal Benefit of Sequential Training

Sequential QCM+DPO training provides only marginal improvements:

- QCM only: 42.65%
- QCM+DPO: 43.37%
- Improvement: +0.72%

The additional computational cost (two training stages) may not justify the marginal gain, especially compared to single-stage DPO-SFT (44.40%).

5.3 Per-Benchmark Analysis

5.3.1 OCRBench

- Baseline: 3.1%
- Best model: ERP DPO-SFT at 53.7%
- All ERP models achieve 52-54% range
- Benchmark-specific training shows minimal improvement (3.4-3.5%)

5.3.2 DocVQA

- Baseline: 7.2%
- Best model: ERP DPO-SFT at 35.1%
- All ERP models achieve 34-35% range
- Training on DocVQA itself achieves only 34.8%

5.3.3 ChartQA

- Baseline: 2.08%
- Best model: ERP DPO-SFT at 44.4%
- Largest improvement: +42.32%
- Training on ChartQA itself achieves only 2.56%

5.4 Training Efficiency

Training characteristics from 1,000-sample experiments:

- Training time: ~9.5 minutes per epoch (900 training samples)
- Training loss reduction: 11.25 $\rightarrow$ 0.0012 (99.99%)
- Sanity check accuracy: 98.4% (on training data)
- GPU memory: Manageable with 4-bit quantization + LoRA

6 Analysis and Discussion

6.1 Why Does ERP Training Succeed?

We hypothesize several factors contributing to ERP training success:

6.1.1 Data Quality and Diversity

ERP datasets may contain more diverse document types, linguistic patterns, and visual layouts representative of real enterprise scenarios. Benchmark datasets, while well-curated, may exhibit narrower distributions.

6.1.2 Task Alignment

ERP training methods (QCM, DPO) align more closely with real-world deployment requirements: understanding document context, preferring high-quality responses, and handling diverse input formats.

6.1.3 Preference Learning

DPO-based methods explicitly learn to distinguish good from bad responses, potentially developing more robust decision boundaries than standard supervised learning.

6.2 Why Does Benchmark Training Fail?

The poor performance of benchmark-specific training reveals several issues:

6.2.1 Overfitting to Benchmark-Specific Patterns

Training exclusively on a single benchmark may cause the model to overfit to that benchmark’s specific characteristics (image styles, question formats, answer distributions) at the expense of general capabilities.

6.2.2 Catastrophic Forgetting

Fine-tuning on narrow task distributions can cause catastrophic forgetting of pre-trained capabilities, as evidenced by below-baseline performance on some benchmarks.

6.2.3 Limited Transfer Learning

Except for DocVQA (which shows strong transfer), most benchmark-specific models fail to generalize. This suggests that benchmark datasets may not capture sufficiently diverse visual-linguistic patterns.

6.3 The DocVQA Anomaly

DocVQA training produces notably different results:

- Competitive with best ERP methods (43.44% vs 44.40%)
- Strong transfer to OCRBench (53.4%) and ChartQA (42.12%)
- Only benchmark-specific model approaching ERP performance

This suggests that DocVQA’s dataset characteristics—diverse document types, complex layouts, varied text recognition requirements—provide a stronger foundation for transfer learning than other benchmarks.

6.4 DPO vs. SFT: A Nuanced Picture

Our results show that DPO-based methods excel, but with nuance:

- DPO-SFT (44.40%) outperforms pure DPO method (43.34%)
- This suggests data quality matters more than training objective
- Preference pairs in DPO datasets may be higher quality than standard SFT data
- Pure DPO’s contrastive loss doesn’t provide additional benefits over simply training on “chosen” responses

6.5 Computational Cost-Benefit Analysis

Table 3: Training cost vs. performance trade-offs

Strategy	Stages	Training Time	Avg. Acc.
Baseline	0	0 min	4.13%
OCRBench SFT	1	~30 min	3.18%
ChartQA SFT	1	~30 min	4.39%
DocVQA SFT	1	~30 min	43.44%
ERP QCM	1	~30 min	42.65%
ERP DPO-SFT	1	~30 min	44.40%
ERP DPO	1	~45 min	43.34%
ERP QCM+DPO	2	~75 min	43.37%

ERP DPO-SFT provides the best performance with single-stage training, making it the most cost-effective approach.

6.6 Limitations

Model Size: We evaluate only SmolVLM-500M. Results may differ for larger models with greater capacity.

Training Scale: Limited to 1,000 samples per benchmark. Larger training sets might alter relative performance.

TextVQA Evaluation: TextVQA results are sparse/missing for most models, limiting conclusions about scene-text capabilities.

Hyperparameter Optimization: We use fixed hyperparameters across experiments. Per-method optimization might improve results.

Dataset Composition: ERP dataset details are limited. More transparency would enable better analysis.

7 Conclusion

This paper presents a systematic evaluation of fine-tuning strategies for vision-language models in enterprise contexts. Through comprehensive experiments comparing benchmark-specific and ERP training methods, we draw several key conclusions:

1. **ERP training significantly outperforms benchmark-specific training**, achieving 44.40% average accuracy compared to 4.13% baseline—a 40.27% improvement.
2. **Benchmark-specific training shows poor generalization**, with most models failing to transfer knowledge across benchmarks. The notable exception is DocVQA, whose diverse document types enable better transfer.

3. **DPO-based methods excel**, with DPO-SFT achieving the best performance. Surprisingly, training on "chosen" responses with SFT outperforms pure DPO method, suggesting data quality matters more than training objective.
4. **Sequential training provides marginal benefits** insufficient to justify additional computational costs.
5. **Training validation is critical**: Our 98.4% sanity check accuracy confirms that proper label masking and training configuration enable effective learning.

7.1 Practical Recommendations

For practitioners deploying VLMs in enterprise settings:

- **Prioritize ERP-style training** over benchmark-specific fine-tuning
- **Use DPO-SFT** (training on high-quality preference pairs with SFT) for best single-stage performance
- **Consider DocVQA** as a fallback if ERP data is unavailable—it shows strong transfer capabilities
- **Avoid single-benchmark overfitting**—diversify training data to maintain general capabilities
- **Implement proper label masking**—critical for effective VLM fine-tuning
- **Use LoRA with 4-bit quantization**—enables efficient training with 0.81% trainable parameters

7.2 Future Work

Several directions warrant further investigation:

- **Larger-scale experiments**: Evaluate with larger models (7B+) and more training samples
- **Multi-task training**: Combine multiple benchmarks and ERP data simultaneously
- **Continual learning**: Investigate methods to prevent catastrophic forgetting
- **ERP dataset analysis**: Deeper understanding of what makes ERP data effective
- **Hybrid approaches**: Combine strengths of DocVQA's diversity with ERP's task alignment
- **Prompt engineering**: Evaluate whether prompt-based adaptation can match fine-tuning

Our systematic evaluation provides a foundation for effective VLM adaptation in enterprise environments, demonstrating that thoughtful training strategy selection yields substantial performance improvements over naive benchmark fine-tuning.

Acknowledgments

We thank the developers of SmolVLM, HuggingFace Transformers, and PEFT libraries for their excellent open-source tools.

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